FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS – GRADE 10 | SUB-STRAND 1.3: QUADRATIC EXPRESSIONS AND EQUATIONS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This Final Explanation document is your opportunity to show everything you have learned across the 8 lessons in this unit. You are writing for an audience that has NOT yet studied quadratic expressions and equations — imagine explaining to a fellow student who missed the whole unit. Your explanation must directly answer the driving question: "How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems — from the arc of a thrown ball on our school field to the dimensions of a farmer's shamba in Kericho?"

READ BEFORE YOU WRITE:

- Use clear, precise mathematical language — define every key term the first time you use it.
- Connect EVERY section back to the anchoring phenomenon (the shot put throw on sports day) OR the Kericho farmer's shamba, or both.
- Show your working — do not just state answers. Explain WHY each step is done, not just HOW.
- Write in full sentences and paragraphs. Bullet points alone are not sufficient.
- You must include at least ONE fully worked numerical example in each section.
- Your explanation should be detailed enough that a student who reads it could solve a new quadratic problem they have never seen before.

Section 1: What Is a Quadratic Expression — and Why Does the Shot Put Curve?	
Explain what a quadratic expression is and how it is different from a linear expression. Use the shot put phenomenon to motivate WHY quadratic expressions matter. In your explanation: (a) define a quadratic expression in the form $ax^2 + bx + c$, identifying what a, b, and c represent and why $a \neq 0$; (b) explain how expanding brackets produces a	When the student athlete on sports day in Nairobi threw the shot put, her classmate noticed the ball traced a smooth curve — not a straight line. A straight-line path can be described by a linear expression like $2x + 3$, where the highest power of x is 1. But the shot put's curved path cannot be captured that way. To describe a curve that rises, reaches a peak, and falls again, we need a new kind of expression — one where x is raised to the power of 2. This is called a quadratic expression. A quadratic expression has the general form $ax^2 + bx + c$, where a, b, and c are constants and $a \neq 0$. The term ax^2 is what gives the expression its curved shape; if a were zero, the x^2 term would disappear and we would be left with a straight line again. For the shot put, a is negative ($a < 0$) because the ball rises and then falls — the parabola opens downward. Quadratic expressions are often formed by expanding (multiplying out) two brackets. For example, imagine a rectangular shamba

quadratic expression; (c) show ONE fully worked example of forming and expanding a quadratic expression from a real context (e.g., the height of the shot put or the area of a shamba). Make sure a reader who has never seen a quadratic expression before would understand what makes it 'quadratic.'	in Kericho where one side is $(x + 3)$ metres and the other side is $(x + 5)$ metres. The area of the shamba is length × width: Area $= (x + 3)(x + 5)$ Expanding using the distributive law (FOIL): $= x \cdot x + x \cdot 5 + 3 \cdot x + 3 \cdot 5$ $= x^2 + 5x + 3x + 15$ $= x^2 + 8x + 15$ Here, $a = 1$, $b = 8$, and $c = 15$. Notice the highest power of x is 2 — that single fact is what makes this expression quadratic. Whether we are calculating the area of a farmer's shamba or modelling the arc of a thrown ball, the quadratic expression $ax^2 + bx + c$ is the mathematical tool that captures the relationship.

Section 2: Factorising Quadratic Expressions — Reverse-Engineering the Curve	
Explain what factorisation means and why it is the reverse of expansion. Describe and demonstrate at least TWO factorisation methods covered in the unit (e.g., finding two numbers that multiply to ac and add to b; factorising a perfect square trinomial; difference of two squares). For each method: (a) state when you would use it; (b) show a fully worked example with clear steps; (c) connect the method back to the shot put or shamba context. Explain in your own words why factorisation is a powerful tool — what does it reveal about a quadratic expression that the expanded form does not show directly?	Factorisation is the process of rewriting a quadratic expression as a product of two brackets — it is the exact reverse of expansion. If expanding is like multiplying out to get $ax^2 + bx + c$, then factorising is like 'undoing' that multiplication to recover the two original brackets. This matters enormously for the shot put problem: once we can factorise the height expression, we can immediately read off the values of x where the ball is at ground level (height = 0). METHOD 1 — Factorising when $a = 1$ (find two numbers that multiply to c and add to b): Suppose the height h (in metres) of the shot put above the ground is modelled by: $h = x^2 - 7x + 10$ where x is the horizontal distance in metres. To factorise, we need two numbers that multiply to $+10$ and add to -7. Testing pairs: $(-2) \times (-5) = +10$ ✓ and $(-2) + (-5) = -7$ ✓. Therefore: $x^2 - 7x + 10 = (x - 2)(x - 5)$ This tells us the ball is at ground level when $x = 2$ or $x = 5$ — giving us the landing distance without any further calculation. METHOD 2 — Difference of Two Squares ($a^2 - b^2 = (a + b)(a - b)$): If a section of the shamba's boundary can be expressed as $x^2 - 16$, we recognise this as a difference of two squares ($x^2 - 4^2$): $x^2 - 16 = (x + 4)(x - 4)$ This method applies whenever the expression has two perfect-square terms separated by a minus sign and no middle term. METHOD 3 — Factorising when $a \neq 1$ (splitting the middle term): Consider $2x^2 + 7x + 3$. Here $a = 2$, $b = 7$, $c = 3$, so $ac = 6$. We need two numbers that multiply to 6 and add to 7: they are 6 and 1. Split the middle term: $2x^2 + 6x + x + 3$ $= 2x(x + 3) + 1(x + 3)$ $= (2x + 1)(x + 3)$

Factorisation is powerful because it transforms the expression from a single 'chunk' ($ax^2 + bx + c$) into a product of simpler parts. Just as you can see the individual ingredients of ugali (maize flour and water) once you understand the recipe, factorisation lets you see the 'ingredients' of the quadratic — and those ingredients directly reveal where the curve crosses the x-axis.

Section 3: Forming and Solving Quadratic Equations — Finding Where the Ball Lands

Explain the difference between a quadratic EXPRESSION and a quadratic EQUATION. Describe how to form a quadratic equation from a real-world context. Then demonstrate how to solve a quadratic equation using: (a) the factorisation method; (b) the quadratic formula. For each method, show a complete worked example using a context from the unit (the shot put trajectory OR the shamba dimensions). Clearly state what the solutions mean in the real-world context — do not just find x; interpret x. Also explain: what does it mean if a quadratic equation has two solutions? One solution? No real solutions?

So far we have been working with quadratic EXPRESSIONS — mathematical phrases like $x^2 + 5x + 6$ that can be simplified or factorised but are not asking us to find a specific value of x. A quadratic EQUATION sets the expression equal to a value (most often zero) and asks: 'For what value(s) of x is this true?' For example:

$x^2 + 5x + 6 = 0$

The addition of '= 0' turns the expression into an equation we can solve.

FORMING A QUADRATIC EQUATION FROM CONTEXT:

A farmer in Kericho wants to fence a rectangular shamba. She has 20 metres of fencing and wants the area to be 24 m². Let the width be x metres; then the length is $(10 - x)$ metres (since $2l + 2w = 20 \rightarrow l + w = 10$). Setting up the area equation:

$x(10 - x) = 24$

$10x - x^2 = 24$

Rearranging: $x^2 - 10x + 24 = 0$

This is our quadratic equation.

METHOD A — Solving by Factorisation:

$x^2 - 10x + 24 = 0$

Find two numbers that multiply to 24 and add to −10: (−4) and (−6).

$(x - 4)(x - 6) = 0$

Using the Zero Product Property: either $(x - 4) = 0 \rightarrow x = 4$, or $(x - 6) = 0 \rightarrow x = 6$.

Interpretation: The shamba can be 4 m × 6 m OR 6 m × 4 m — both describe the same rectangle, just with length and width swapped. Both solutions make physical sense. ✓

METHOD B — Solving using the Quadratic Formula:

For any equation $ax^2 + bx + c = 0$, the solutions are given by:

$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$

Using our equation $x^2 - 10x + 24 = 0$ ($a=1$, $b=-10$, $c=24$):

$x = (10 \pm \sqrt{100 - 96}) / 2 = (10 \pm \sqrt{4}) / 2 = (10 \pm 2) / 2$

$x = 6$ or $x = 4$ ✓ — the same answers, confirming both methods work.

The expression under the square root sign ($b^2 - 4ac$) is called the DISCRIMINANT. If it is positive, there are TWO real solutions (the ball lands at two measurable points). If it equals zero, there is exactly ONE solution (the ball just grazes the ground). If it is negative, there are NO real solutions — meaning the scenario described is physically impossible (e.g., the shamba area requested is larger than the fencing can enclose).

Section 4: Connecting All Three Skills — Answering the Driving Question	
Now bring everything together to directly and fully answer the driving question. Walk the reader through a complete, end-to-end problem that requires ALL THREE skills: (1) forming a quadratic expression, (2) factorising it, and (3) solving the resulting equation. The problem MUST connect to either the shot put phenomenon or the Kericho shamba (or both). After solving, reflect: How does quadratic mathematics let us PREDICT something before it happens (e.g., where the ball will land before it lands)? How does it help us EXPLAIN the curved shape we observed? How does it help us SOLVE practical problems like the shamba? Write this section as if you are speaking directly to the student who first asked 'what kind of rule could describe this curve?'	Remember the moment on sports day when the class asked: 'What mathematical rule describes that curve — and could it tell us where the ball lands before it lands?' You now have all the tools to answer that question completely. Here is a complete problem that connects all three skills: During sports day at Nairobi's school, the height h (in metres) of the shot put above the ground is modelled by the expression $-x^2 + 6x$, where x is the horizontal distance in metres from the throwing point. The class wants to know: (a) At what horizontal distances is the ball at ground level? (b) What is the maximum height reached? STEP 1 — Form the equation: The ball is at ground level when $h = 0$: $-x^2 + 6x = 0$ STEP 2 — Factorise: Factor out $-x$ from both terms: $-x(x - 6) = 0$ STEP 3 — Solve: Using the Zero Product Property: $-x = 0 \rightarrow x = 0$ (the throwing point) $x - 6 = 0 \rightarrow x = 6$ (the landing point) INTERPRETATION: The ball leaves the ground at $x = 0$ m and lands at $x = 6$ m. Without catching the ball or waiting for it to land, your class can PREDICT the landing point algebraically — exactly what the teacher challenged you to do. For the maximum height: by symmetry, the ball reaches its peak halfway between $x = 0$ and $x = 6$, i.e., at $x = 3$ m. Substituting: $h = -(3)^2 + 6(3) = -9 + 18 = 9$ metres. So the shot put reaches a maximum height of 9 metres at a horizontal distance of 3 metres — all predicted from the equation, before even watching the full video. This is the power of quadratic mathematics: it EXPLAINS why the path curves (because of the x^2 term), it PREDICTS outcomes (landing distance, peak height) before they are observed, and it SOLVES practical problems (like determining shamba dimensions that satisfy an area requirement). The smooth, symmetrical arch filmed on sports day in Nairobi is not random — it is a parabola, and now you have the algebra to decode it completely.

Section 5: Reflection — Limitations, Extensions, and What You Now Wonder	
Reflect critically on the mathematical models you have used in this unit. Address the following: (a) What assumptions did the quadratic model make about the shot put's path? Are these assumptions perfectly accurate in real life? What factors might cause the real path to differ slightly from the model? (b) Give ONE new real-life situation in Kenya (different from the shot put and the shamba) where you think a quadratic model would be useful, and briefly explain why. (c) What new questions does this unit raise for you that you have NOT yet been able to answer? Add at least one question to your personal Driving Question Board. This section shows that you think like a mathematician — not just someone who can follow steps, but someone who questions, extends, and wonders.	Every mathematical model is a simplification of reality, and the quadratic model for the shot put is no exception. Our model assumed that the only force acting on the ball after it is thrown is gravity — it ignored air resistance, wind on the Nairobi school field, and the spin of the ball. In a real throw, these factors mean the actual path will deviate slightly from a perfect parabola. The model also assumed the ball was released at ground level ($h = 0$ when $x = 0$), but in practice the athlete releases the shot put from shoulder height, so a more accurate model might start at $h = 1.5$ m. Despite these limitations, the quadratic model gives a very close approximation — close enough to be genuinely useful for predicting landing distance. A NEW KENYAN CONTEXT where quadratic mathematics applies: Imagine a matatu operator in Nairobi who wants to set the fare price to maximise daily revenue. If she charges x shillings per passenger, her experience shows that the number of passengers per day is $(200 - 2x)$. Her daily revenue R is: $R = x(200 - 2x) = 200x - 2x^2$ This is a quadratic expression in x — and finding the fare that maximises revenue requires exactly the skills from this unit (identifying the vertex of the parabola). The same mathematics that describes the shot put also helps a business owner make better decisions. MY DRIVING QUESTION BOARD ADDITION: This unit answered many of my early questions, but it opened new ones. Here is a question I am adding to my Driving Question Board: 'Our model gave us the landing distance, but can we also use quadratic equations to find the ANGLE at which the ball was thrown to achieve the greatest possible distance — and is that the same angle for every throw, regardless of the athlete's strength?' Thinking like a mathematician means recognising that every answer is also a doorway to a deeper question. The parabolic throw on sports day was just the beginning.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Conceptual Understanding of Quadratic Expressions and Equations	Accurately and clearly defines quadratic expressions ($ax^2 + bx + c$, $a \neq 0$) and equations, consistently distinguishing them from linear forms. Explains the role of each coefficient with precision. All mathematical statements are correct and complete.	Correctly defines quadratic expressions and equations in most instances. Minor gaps in explaining the significance of $a \neq 0$ or the role of individual coefficients, but no major conceptual errors.	Shows partial understanding of what makes an expression quadratic. May confuse expressions with equations, or omit the condition $a \neq 0$. Key definitions are incomplete or contain errors.
Procedural Accuracy and Worked Examples	All worked examples are fully correct with every step shown and clearly labelled. Uses at least two factorisation methods	Most worked examples are correct with steps shown. One or two minor arithmetic or algebraic slips that do not derail the	Worked examples are incomplete or contain significant errors in expansion, factorisation, or application of the

	and the quadratic formula correctly. Arithmetic and algebraic manipulations are error-free throughout.	overall solution. Both factorisation and the quadratic formula are attempted with general accuracy.	quadratic formula. Steps are missing or the logic of the procedure is unclear.
Connection to Real-World Context and Phenomenon	Every section is explicitly and meaningfully connected to the shot put phenomenon and/or the Kericho shamba. Solutions are interpreted in real-world terms (units stated, meaning of x explained). The driving question is answered directly and completely in Section 4.	Most sections reference the real-world contexts. Solutions are interpreted but interpretations may lack full detail (e.g., units occasionally missing). The driving question is addressed but the answer could be more complete or direct.	Real-world connections are superficial or limited to one or two sections. Mathematical solutions are stated without interpretation. The driving question is not clearly or directly answered.
Communication: Clarity, Structure, and Mathematical Language	Explanation is written in clear, full sentences with accurate mathematical vocabulary used consistently (e.g., coefficient, factorise, discriminant, roots, parabola). Each section flows logically. A student who missed the unit could follow and learn from this explanation independently.	Explanation is generally clear and mostly uses correct mathematical vocabulary. A few terms may be used imprecisely or undefined. The structure is logical but some transitions between ideas could be clearer.	Explanation is difficult to follow in places. Mathematical vocabulary is limited, missing, or incorrectly used. Sections lack clear structure or the logical flow between steps is not apparent to a new reader.
Critical Reflection and Mathematical Thinking (Section 5)	Identifies specific, realistic limitations of the quadratic model with justification. Proposes a novel, well-reasoned Kenyan context for applying quadratic mathematics. Poses a genuine, mathematically meaningful new question that extends beyond the unit content.	Acknowledges at least one limitation of the model. Proposes a relevant new Kenyan context, though the explanation of why quadratics apply may lack depth. New question is relevant but may not extend significantly beyond unit content.	Reflection is superficial or generic (e.g., 'the model might not be perfect'). New context is either absent, not Kenyan, or the quadratic connection is not explained. No meaningful new question is posed.